

Лабораторная работа №3

Операционные системы

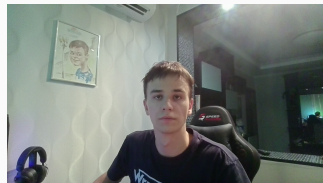
Савкин Д. В.

02 сентября 2026

Российский университет дружбы народов, Москва, Россия

Информация

- Савкин Дмитрий Вадимович
- РУДН, группа НПИбд-02-25
- Тема: оформление отчётов в Markdown



Цель работы

- Научиться оформлять отчёты с помощью Markdown

Задание

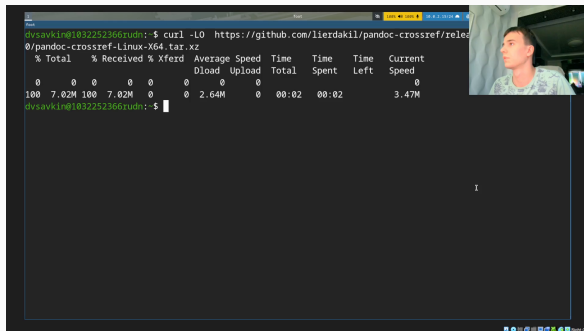
Задание

- Сделать отчёт по Лабораторной работе №2 в формате Markdown
- Предоставить отчёт в форматах pdf, docx и md

Выполнение работы

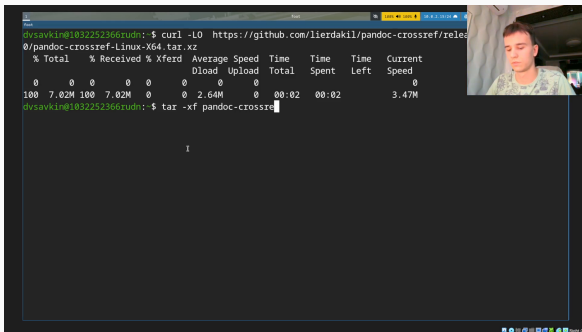
Скачивание pandoc-crossref

Скачиваем сборку pandoc-crossref, соответствующую версии pandoc.



```
dvsavkin@1032252366rudn:~$ curl -LO https://github.com/lierdakil/pandoc-crossref/releases/download/v0.10.0/pandoc-crossref-Linux-X64.tar.xz
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left   Speed
  0     0    0     0    0     0      0      0      0     0      0      0      0
100 7.02M 100 7.02M    0     0  2.64M      0    00:02    00:02    00:02  3.47M
dvsavkin@1032252366rudn:~$
```

Распаковываем архив: `tar -xf`.



A terminal window showing the process of downloading and extracting a tar archive. The user is in a shell with the prompt `dvsavkin@1032252366rudn:~$`. They run `curl -LO https://github.com/lierdakil/pandoc-crossref/releases/download/0/pandoc-crossref-Linux-X64.tar.xz`. The output shows a progress bar and a table of statistics. Then they run `tar -xf pandoc-crossref-Linux-X64.tar.xz`. The output shows the extraction of files.

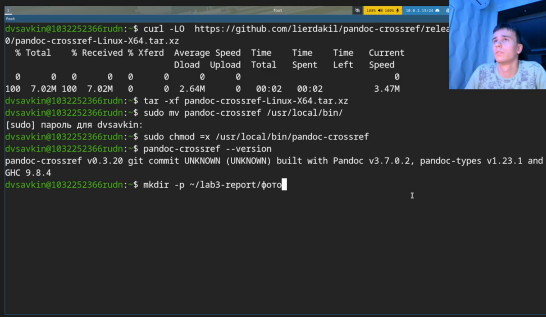
```
dvsavkin@1032252366rudn:~$ curl -LO https://github.com/lierdakil/pandoc-crossref/releases/download/0/pandoc-crossref-Linux-X64.tar.xz
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left   Speed
  0     0     0      0     0      0     0      0      0     0      0      0      0
100 7.02M 100 7.02M    0     0  2.64M      0  00:02  00:02   00:02  3.47M

dvsavkin@1032252366rudn:~$ tar -xf pandoc-crossref-Linux-X64.tar.xz
I
```

Переносим программу в /usr/local/bin, даём права, проверяем версию.

```
dvysavkin@1032252366rudn:~$ curl -LO https://github.com/lierdakil/pandoc-crossref/releases/download/v0.3.20/pandoc-crossref-Linux-X64.tar.xz
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left   Speed
  0     0    0     0    0     0      0      0  00:01  00:01  00:01  0
100  7.02M 100  7.02M    0     0  3.51M      0  00:01  00:01  00:01 2.17M
dvysavkin@1032252366rudn:~$ tar -xvf pandoc-crossref-Linux-X64.tar.xz
dvysavkin@1032252366rudn:~$ sudo mv pandoc-crossref /usr/local/bin/
dvysavkin@1032252366rudn:~$ sudo chmod 755 /usr/local/bin/pandoc-crossref
pandocdvysavkin@1032252366rudn:~$ ndoc-crossref --version
bash: ndoc-crossref: command not found
dvysavkin@1032252366rudn:~$ pandoc-crossref --version
pandoc-crossref v0.3.20 git commit UNKNOWN (UNKNOWN) built with Pandoc v3.7.0.2, pandoc-types v1.23.1 and
GHC 9.8.4
dvysavkin@1032252366rudn:~$
```

Создаём рабочий каталог для отчёта с подпапкой для скриншотов.



```

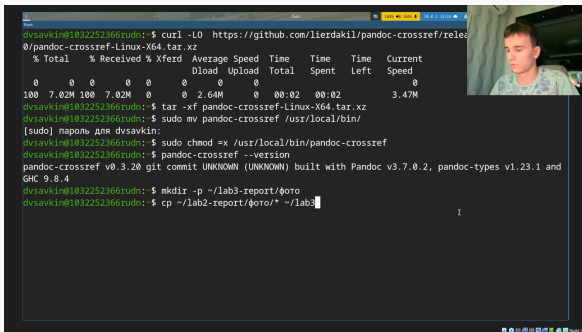
dvsavkin@1032252366rudn:~$ curl -LO https://github.com/lierdakil/pandoc-crossref/releases
0/pandoc-crossref-Linux-X64.tar.xz
% Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
           Dload  Upload   Total   Spent    Left   Speed

  0     0    0     0  0     0     0     0      0  0
100  7.02M 100  7.02M  0     0   2.64M    0  00:02  00:02    3.47M

dvsavkin@1032252366rudn:~$ tar -xf pandoc-crossref-Linux-X64.tar.xz
dvsavkin@1032252366rudn:~$ sudo mv pandoc-crossref /usr/local/bin/
[sudo] пароль для dvsavkin:
dvsavkin@1032252366rudn:~$ sudo chmod =x /usr/local/bin/pandoc-crossref
dvsavkin@1032252366rudn:~$ pandoc-crossref --version
pandoc-crossref v0.3.20 git commit UNKNOWN (UNKNOWN) built with Pandoc v3.7.0.2, pandoc-types v1.23.1 and
GHC 9.8.4
dvsavkin@1032252366rudn:~$ mkdir -p ~/lab3-report/0to0

```

Копируем текст отчёта по Лабораторной работе №2.



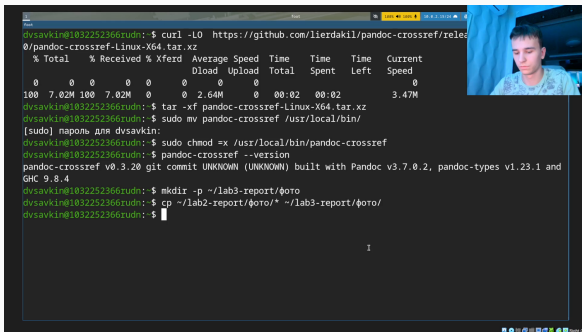
```
dvsavkin@1032252366rudn:~$ curl -LO https://github.com/lierdakil/pandoc-crossref/releases/download/v0.3.20/pandoc-crossref-Linux-X64.tar.xz
dvsavkin@1032252366rudn:~$ tar -xvf pandoc-crossref-Linux-X64.tar.xz
dvsavkin@1032252366rudn:~$ sudo mv pandoc-crossref /usr/local/bin/
[sudo] пароль для dvsavkin:
dvsavkin@1032252366rudn:~$ sudo chmod =x /usr/local/bin/pandoc-crossref
dvsavkin@1032252366rudn:~$ pandoc-crossref --version
pandoc-crossref v0.3.20 git commit UNKNOWN (UNKNOWN) built with Pandoc v3.7.0.2, pandoc-types v1.23.1 and GHC 9.8.4
dvsavkin@1032252366rudn:~$ mkdir -p ~/lab3-report/0oro
dvsavkin@1032252366rudn:~$ cp ~/lab2-report/0oro/* ~/lab3-
```

% Total	% Received	% Xferd	Average Speed	Time Dload	Time Upload	Time Total	Time Spent	Time Left	Current Speed
0	0	0	0	0	0	0	0	0	0
100	7.02M	100	7.02M	0	0	2.64M	00:02	00:02	3.47M

The image shows a terminal window with the above commands and output. A video inset in the top right corner shows a person's face. The terminal output includes a progress table for the download of pandoc-crossref.

Копирование скриншотов

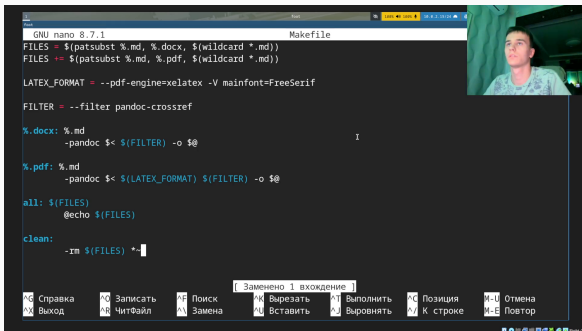
Копируем скриншоты Лабораторной работы №2.



```
dvsavkin@1032252366rudn:~$ curl -LO https://github.com/lierdakil/pandoc-crossref/releases/download/v0.3.20/pandoc-crossref-Linux-X64.tar.xz
dvsavkin@1032252366rudn:~$ tar -xvf pandoc-crossref-Linux-X64.tar.xz
dvsavkin@1032252366rudn:~$ sudo mv pandoc-crossref /usr/local/bin/
[sudo] пароль для dvsavkin:
dvsavkin@1032252366rudn:~$ sudo chmod =x /usr/local/bin/pandoc-crossref
dvsavkin@1032252366rudn:~$ pandoc-crossref --version
pandoc-crossref v0.3.20 git commit UNKNOWN (UNKNOWN) built with Pandoc v3.7.0.2, pandoc-types v1.23.1 and GHC 9.8.4
dvsavkin@1032252366rudn:~$ mkdir -p ~/lab3-report/0000
dvsavkin@1032252366rudn:~$ cp ~/lab2-report/0000/* ~/lab3-report/0000/
dvsavkin@1032252366rudn:~$
```

The terminal window shows the process of downloading the pandoc-crossref binary from GitHub, extracting it, moving it to the system's bin directory, setting permissions, and then using it to create a directory and copy files. A small video inset in the top right corner shows a person's face.

Создаём Makefile: xelatex + FreeSerif + фильтр pandoc-crossref.



```
GNU nano 8.7.1 Makefile
FILES = $(patsubst %.md, %.docx, $(wildcard *.md))
FILES += $(patsubst %.md, %.pdf, $(wildcard *.md))

LATEX_FORMAT = --pdf-engine=xelatex -V mainfont=FreeSerif

FILTER = --filter pandoc-crossref

%.docx: %.md
    -pandoc $< $(FILTER) -o $@

%.pdf: %.md
    -pandoc $< $(LATEX_FORMAT) $(FILTER) -o $@

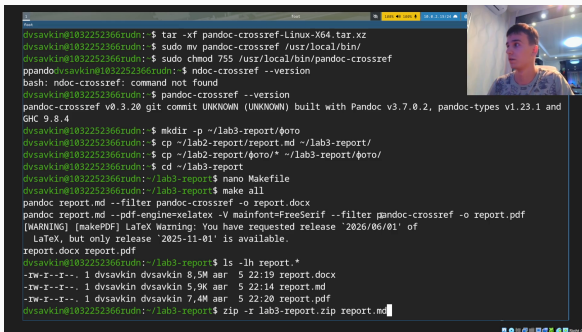
all: $(FILES)
    @echo $(FILES)

clean:
    -rm $(FILES) +~
```

Заменено 1 вхождение

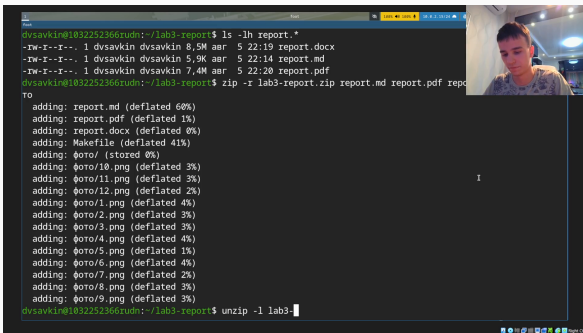
⌘ Справка ⌘ Записать ⌘ Поиск ⌘ Вырезать ⌘ Выполнить ⌘ Позиция ⌘ Отмена
⌘ Выход ⌘ ЧитФайл ⌘ Замена ⌘ Вставить ⌘ Выровнять ⌘ К строке ⌘ Повтор

Запускаем `make all` — собираются `report.docx` и `report.pdf`.



```
dvsavkin@1032252366rudn:~$ tar -xvf pandoc-crossref-Linux-X64.tar.xz
dvsavkin@1032252366rudn:~$ sudo mv pandoc-crossref /usr/local/bin/
dvsavkin@1032252366rudn:~$ sudo chmod 755 /usr/local/bin/pandoc-crossref
ppando@dvsavkin@1032252366rudn:~$ ndoc-crossref --version
bash: ndoc-crossref: command not found
dvsavkin@1032252366rudn:~$ pandoc-crossref --version
pandoc-crossref v0.3.20 git commit UNKNOWN (UNKNOWN) built with Pandoc v3.7.0.2, pandoc-types v1.23.1 and GHC 9.8.4
dvsavkin@1032252366rudn:~$ mkdir -p ~/lab3-report/070
dvsavkin@1032252366rudn:~$ cp ~/lab2-report/report.md ~/lab3-report/
dvsavkin@1032252366rudn:~$ cp ~/lab2-report/070/* ~/lab3-report/070/
dvsavkin@1032252366rudn:~$ cd ~/lab3-report
dvsavkin@1032252366rudn:~/lab3-report$ nano Makefile
dvsavkin@1032252366rudn:~/lab3-report$ make all
pandoc report.md --filter pandoc-crossref -o report.docx
pandoc report.md --pdf-engine=xelatex -V mainfont=FreeSerif --filter pandoc-crossref -o report.pdf
[WARNING] [makePDF] LaTeX Warning: You have requested release '2026/06/01' of
LaTeX, but only release '2025-11-01' is available.
report.docx report.pdf
dvsavkin@1032252366rudn:~/lab3-report$ ls -lh report.*
-rw-r--r-- 1 dvsavkin dvsavkin 8,5M аэр 5 22:19 report.docx
-rw-r--r-- 1 dvsavkin dvsavkin 5,9K аэр 5 22:14 report.md
-rw-r--r-- 1 dvsavkin dvsavkin 7,4M аэр 5 22:20 report.pdf
dvsavkin@1032252366rudn:~/lab3-report$ zip -r lab3-report.zip report.md
```

Упаковываем всё в архив и проверяем содержимое.

A terminal window with a dark background and green text. The prompt is 'dvsavkin@1032252366rudn:~/lab3-report\$'. The first command is 'ls -lh report.*', which lists three files: 'report.docx' (8.5M), 'report.md' (5.9K), and 'report.pdf' (7.4M). The second command is 'zip -r lab3-report.zip report.md report.pdf report.docx'. The output shows the progress of adding each file to the archive, including compression percentages. The final command is 'unzip -l lab3-'.

```
dvsavkin@1032252366rudn:~/lab3-report$ ls -lh report.*
-rw-r--r--. 1 dvsavkin dvsavkin 8,5M аэр 5 22:19 report.docx
-rw-r--r--. 1 dvsavkin dvsavkin 5,9K аэр 5 22:14 report.md
-rw-r--r--. 1 dvsavkin dvsavkin 7,4M аэр 5 22:20 report.pdf
dvsavkin@1032252366rudn:~/lab3-report$ zip -r lab3-report.zip report.md report.pdf report.docx
adding: report.md (deflated 60%)
adding: report.pdf (deflated 1%)
adding: report.docx (deflated 0%)
adding: Makefile (deflated 41%)
adding: foto/ (stored 0%)
adding: foto/10.png (deflated 3%)
adding: foto/11.png (deflated 3%)
adding: foto/12.png (deflated 2%)
adding: foto/1.png (deflated 4%)
adding: foto/2.png (deflated 3%)
adding: foto/3.png (deflated 3%)
adding: foto/4.png (deflated 4%)
adding: foto/5.png (deflated 1%)
adding: foto/6.png (deflated 4%)
adding: foto/7.png (deflated 2%)
adding: foto/8.png (deflated 3%)
adding: foto/9.png (deflated 3%)
dvsavkin@1032252366rudn:~/lab3-report$ unzip -l lab3-
```

Выводы

- Изучены основы Markdown, освоена сборка отчётов через Pandoc
- Установлен pandoc-crossref, написан Makefile для сборки в 2 форматах